PREMIUM

Full-Length Mock Test

Mock 03

Life Sciences

GATE XL

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Chemistry (XL-P) — Electrochemistry]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.2 [1 Mark] [EASY] [Biochemistry — Metabolism]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.3 [1 Mark] [EASY] [Botany — Cell Biology]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.4 [1 Mark] [EASY] [Microbiology — Viruses]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.5 [1 Mark] [EASY] [Chemistry (XL-P) — Electrochemistry]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.6 [1 Mark] [EASY] [Biochemistry — Nucleic Acids]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.7 [1 Mark] [EASY] [Botany — Ecology]

DNA double helix was discovered by:

A) Linus Pauling

B) Watson & Crick

C) Rosalind Franklin

D) Gregor Mendel

Q.8 [1 Mark] [EASY] [Microbiology — Bacteria]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.9 [1 Mark] [EASY] [Chemistry (XL-P) — Kinetics]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.10 [1 Mark] [EASY] [Biochemistry — Metabolism]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.11 [1 Mark] [EASY] [Botany — Genetics]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.12 [1 Mark] [EASY] [Microbiology — Bacteria]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.13 [2 Marks] [MODERATE] [Chemistry (XL-P) — Kinetics]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.14 [2 Marks] [MODERATE] [Biochemistry — Proteins]

Which bond links amino acids?

A) Hydrogen bond

B) Peptide bond

C) Phosphodiester

D) Glycosidic

Q.15 [2 Marks] [MODERATE] [Botany — Cell Biology]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.16 [2 Marks] [MODERATE] [Microbiology — Immunology]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.17 [2 Marks] [MODERATE] [Chemistry (XL-P) — Electrochemistry]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.18 [2 Marks] [MODERATE] [Biochemistry — Metabolism]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.19 [2 Marks] [MODERATE] [Botany — Cell Biology]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.20 [2 Marks] [MODERATE] [Microbiology — Immunology]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.21 [2 Marks] [MODERATE] [Chemistry (XL-P) — Atomic Structure]

Van der Waals equation accounts for:

A) Attraction + volume

B) Volume only

C) Attraction only

D) None

Q.22 [2 Marks] [MODERATE] [Biochemistry — Metabolism]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.23 [2 Marks] [MODERATE] [Botany — Ecology]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.24 [2 Marks] [MODERATE] [Microbiology — Viruses]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.25 [2 Marks] [MODERATE] [Chemistry (XL-P) — Chemical Bonding]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.26 [2 Marks] [MODERATE] [Biochemistry — Bioenergetics]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.27 [2 Marks] [MODERATE] [Botany — Cell Biology]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.28 [2 Marks] [MODERATE] [Microbiology — Genetics]

DNA double helix was discovered by:

A) Linus Pauling

B) Watson & Crick

C) Rosalind Franklin

D) Gregor Mendel

Q.29 [2 Marks] [MODERATE] [Chemistry (XL-P) — Surface Chemistry]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.30 [2 Marks] [MODERATE] [Biochemistry — Bioenergetics]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.31 [2 Marks] [MODERATE] [Botany — Taxonomy]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.32 [2 Marks] [MODERATE] [Microbiology — Genetics]

Which bond links amino acids?

A) Hydrogen bond
B) Peptide bond
C) Phosphodiester
D) Glycosidic

Q.33 [2 Marks] [HARD] [Chemistry (XL-P) — Kinetics]

Van der Waals equation accounts for:

A) Attraction + volume
B) Volume only
C) Attraction only
D) None

Q.34 [2 Marks] [HARD] [Biochemistry — Nucleic Acids]

DNA double helix was discovered by:

A) Linus Pauling
B) Watson & Crick
C) Rosalind Franklin
D) Gregor Mendel

Q.35 [2 Marks] [HARD] [Botany — Cell Biology]

Which bond links amino acids?

A) Hydrogen bond

B) Peptide bond

C) Phosphodiester

D) Glycosidic

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Microbiology — Microbial Ecology]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Chemistry (XL-P) — Kinetics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Biochemistry — Metabolism]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Botany — Taxonomy]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Microbiology — Viruses]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Botany — Genetics]

Standard calculation using basic formula:

Answer: ____________________

Q.52 [2 Marks] [HARD] [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Chemistry (XL-P) — Electrochemistry]

Standard calculation using basic formula:

Answer: ____________________

Q.54 [2 Marks] [HARD] [Biochemistry — Proteins]

Standard calculation using basic formula:

Answer: ____________________

Q.55 [2 Marks] [HARD] [Botany — Ecology]

Standard calculation using basic formula:

Answer: ____________________

Q.56 [2 Marks] [HARD] [Microbiology — Microbial Ecology]

Standard calculation using basic formula:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Chemistry (XL-P) — Atomic Structure]

Standard calculation using basic formula:

Answer: ____________________

Q.58 [2 Marks] [HARD] [Biochemistry — Nucleic Acids]

Standard calculation using basic formula:

Answer: ___________________________

Q.59 [2 Marks] [HARD] [Botany — Cell Biology]

Standard calculation using basic formula:

Answer: ______________________

Q.60 [2 Marks] [HARD] [Microbiology — Viruses]

Standard calculation using basic formula:

Answer: ______________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	B	**Q.3:**	A
Q.4:	B	**Q.5:**	B	**Q.6:**	A
Q.7:	B	**Q.8:**	B	**Q.9:**	A
Q.10:	B	**Q.11:**	B	**Q.12:**	A
Q.13:	B	**Q.14:**	B	**Q.15:**	A
Q.16:	B	**Q.17:**	B	**Q.18:**	A
Q.19:	B	**Q.20:**	B	**Q.21:**	A
Q.22:	B	**Q.23:**	B	**Q.24:**	A
Q.25:	B	**Q.26:**	B	**Q.27:**	A
Q.28:	B	**Q.29:**	B	**Q.30:**	A
Q.31:	B	**Q.32:**	B	**Q.33:**	A
Q.34:	B	**Q.35:**	B	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	10
Q.52:	10	**Q.53:**	10	**Q.54:**	10
Q.55:	10	**Q.56:**	10	**Q.57:**	10
Q.58:	10	**Q.59:**	10	**Q.60:**	10

Detailed Solutions

Question 1
[Chemistry (XL-P) — Electrochemistry]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 2
[Biochemistry — Metabolism]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 3
[Botany — Cell Biology]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 4
[Microbiology — Viruses]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 5
[Chemistry (XL-P) — Electrochemistry]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 6
[Biochemistry — Nucleic Acids]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 7
[Botany — Ecology]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 8
[Microbiology — Bacteria]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 9
[Chemistry (XL-P) — Kinetics]

Van der Waals equation accounts for:

Answer:

(A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 10 *[Biochemistry — Metabolism]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 11 *[Botany — Genetics]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 12 *[Microbiology — Bacteria]*

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 13 *[Chemistry (XL-P) — Kinetics]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 14 *[Biochemistry — Proteins]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 15 *[Botany — Cell Biology]*

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 16 *[Microbiology — Immunology]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 17 *[Chemistry (XL-P) — Electrochemistry]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 18 *[Biochemistry — Metabolism]*

Van der Waals equation accounts for:

Answer:

(A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 19 *[Botany — Cell Biology]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 20 *[Microbiology — Immunology]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 21 *[Chemistry (XL-P) — Atomic Structure]*

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 22 *[Biochemistry — Metabolism]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 23 *[Botany — Ecology]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 24 *[Microbiology — Viruses]*

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 25 *[Chemistry (XL-P) — Chemical Bonding]*

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 26 *[Biochemistry — Bioenergetics]*

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 27 *[Botany — Cell Biology]*

Van der Waals equation accounts for:

Answer:

(A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 28
[Microbiology — Genetics]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 29
[Chemistry (XL-P) — Surface Chemistry]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 30
[Biochemistry — Bioenergetics]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 31
[Botany — Taxonomy]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 32
[Microbiology — Genetics]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 33
[Chemistry (XL-P) — Kinetics]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 34
[Biochemistry — Nucleic Acids]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 35
[Botany — Cell Biology]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 36
[Microbiology — Microbial Ecology]

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37
[Chemistry (XL-P) — Kinetics]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38
[Biochemistry — Metabolism]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39
[Botany — Taxonomy]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40
[Microbiology — Viruses]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41
[Chemistry (XL-P) — Surface Chemistry]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42
[Biochemistry — Metabolism]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43
[Botany — Plant Physiology]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44
[Microbiology — Viruses]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45
[Chemistry (XL-P) — Electrochemistry]

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Biochemistry — Bioenergetics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Botany — Cell Biology]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Microbiology — Microbial Ecology]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Chemistry (XL-P) — Kinetics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Biochemistry — Enzymes]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Botany — Genetics]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 *[Microbiology — Bacteria]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 *[Chemistry (XL-P) — Electrochemistry]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 *[Biochemistry — Proteins]*

Standard calculation using basic formula:

Answer:

(10)

Apply the appropriate formula for the given data.

Question 55 *[Botany — Ecology]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 *[Microbiology — Microbial Ecology]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 *[Chemistry (XL-P) — Atomic Structure]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 *[Biochemistry — Nucleic Acids]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 *[Botany — Cell Biology]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 *[Microbiology — Viruses]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
